EXAMINATION CONFIDENTIAL

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বিএআইইউএসটি), কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Department of CSE

Level-2, Term-II

Term Final Examination, Fall 2024

Course Code: GED221

Course Title: Engineering Economics **Time:3 hrs**

Marks:150

Notes:

a. The figure on the right of each question indicates marks for the respective question.
b. There are two parts in this question.
Part – A: Total seven questions, answer **any five.**
Part – B: Total two questions, answer **both.**

Part A(5*20=100)

1. a) Explain the relationship between AC and MC. **[8]**
b) What do you understand by elasticity of demand? Explain Different types of price elasticity of demand. **[12]**

2. a) Define marginal utility. Explain the Law of Diminishing Marginal Product. **[8]**
b) Analyze economies of scale and diseconomies of scale with graph. **[4]**
c) Explain criticisms of capitalism. **[8]**

3. a) Explain the reasons of inflation in an economy. What are the impacts of inflation? **[12]**
b) Explain differences between cost push and demand pull inflation with graph. **[8]**

4. a) Explain different types of unemployment. What are the reasons of unemployment problem in Bangladesh? **[13]**
b) Differentiate between fixed costs and variable costs with graph. **[7]**

5. a) Draw the circular flow diagram linking households, business firms, and the international sector. Use the diagram to explain the effects of a decision by the household sector to increase saving. **[12]**
b) Analyze real GDP and nominal GDP with a quantitative example. **[8]**

6. a) Define LDCs.Explain the positive sides for Bangladesh becoming developing country. What are the challenges? [13]
   b) Explain the conditions of price discrimination with definition. [7]

7. a) Define monopolistic competition with examples. Analyze how a firm under monopolistic competition attains equilibrium in the short run. [12]
   b) Differentiate between normal profit and super normal profit with a quantitative example. [8]

Part B(25+25=50)

8. Suppose the demand and supply curves of salt are given by: 25

Qd=1000-P

Qs=700+2P

   a) Find the equilibrium price and quantity.
   b) Now suppose that the price of an input used to produce salt increased so that the new supply curve is Qs=400+2P. How does the equilibrium price and quantity change?
   c) Suppose that the government has imposed a tax of tk. 3 per unit of sale. How does it affect the equilibrium price and quantity?
   d) Show all the results/ effects through graphs.

9. Use following information to find the value of **GDP; GNP; NNP; NI; PI; DPI.** 25

Consumption	$600
Gross Investment	$100
Government Spending	$200
Exports	$100
Imports	$150
Income earned but not received	$20
Income received but not earned	$30
Personal taxes	$200

Depreciation	$230
Receipts of factor income from rest of the world	$50
Payments of factor income to the rest of the world	$50
Indirect Business taxes	$90

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

EXAMINATION CONFIDENTIAL

Term Final Examination, FALL 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Each question carries **30** marks.
b. Answer any **5 (five)** questions out of **7 (Seven)**
c. The figure on the right of each question indicates marks for the respective question.
d. Use a different answer script for each part.
e. Course Learning Outcomes are:

CO1: Remember and understand the number system and Boolean algebra and basic properties of Boolean algebra to simplify simple Boolean functions.

CO2: Understanding and applying the tabulation and Karnaugh map methods for simplifying combinational circuits.

CO3: Identify the basic sequential logic components: SR Latch, Different Flip-Flops and their usage and able to analyze sequential logic circuits.

CO4: Design and develop different digital systems like shifters, counters, registers by presenting in front of the class.

Part-A

1. a. A circuit has 3 inputs (A, B, C) and 1 output (F). The output is **1** if the majority of inputs are 1. **[CO1]** — 10
   i) Write the truth table.
   ii) Derive the SOP expression and implement it with basic gates (AND, OR, NOT).

   b. Proof that "***NAND***" & "**NOR**" are Universal gate. **[CO1]** — 10

   c. What are the primary differences between 1's complement and 2's complement? **[CO1]** — 10

2. a. Explain different types of adder & subtractor. **[CO1]** — 10

   b. Draw a 3:8-line decoder and 8:3-line encoder with proper truth table. **[CO1]** — 10

   c. Obtain an 8X1 multiplexer with a dual 4X1 multiplexers having separate enable inputs but common selection lines. Use a block diagram construction. **[CO2]** — 10

3. a. Design a converter that translates BCD to excess-3 code, and then illustrate your solution with the corresponding logic diagram. [**CO2**] 15

Or

Explain clocked RS flipflop with proper diagram. [**CO3**]

b. Design a combinational circuit using a ROM. The circuit accepts a 3-bit number and generates an output binary number equal to the square of the input number. [**CO1**] 15

Or

What do you mean by Multiplexer & Demultiplexer? Design a 6:1-line multiplexer & 1:5-line de-multiplexer. [**CO1**]

Part-B

4. a. Draw the circuit diagram of a 2-bit parallel adder and show the calculation if A=01 & B=10. [**CO1**] 15

b. Draw the circuit diagram of a 2-bit binary multiplier and show the calculation if A=11 & B=10. [**CO1**] 15

5. a. Mention few applications of ROM. Implement the following functions with a ROM. [**CO2**] 10

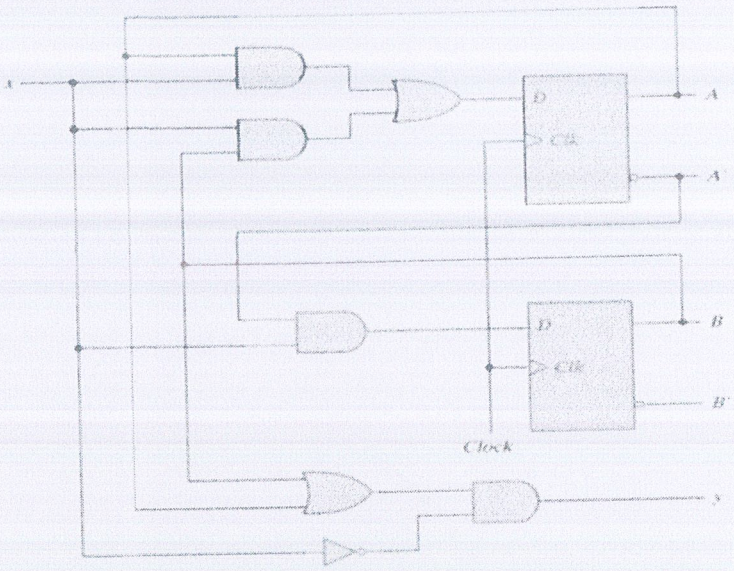
$$F_1(A, B, C) = \sum (1, 3, 7)$$
$$F_2(A, B, C) = \sum (0, 2, 4)$$

b. Design and implement the following function using a 4-to-1 multiplexer along with any necessary external logic gates. [**CO2**] 10

$$F(A, B, C, D) = \sum (1, 3, 4, 11, 12, 13, 14, 15).$$

c. Construct a 5X32 decoder with four 3X8 decoder DMUX and a 2X4 decoder. Use a block diagram? [**CO2**] 10

6. a. Derive state table & state diagram for the following sequential circuit. [**CO3**] 10



b. Design a 3-bit synchronous up/down counter using flip-flops. [**CO3**] 10

Or

Draw the JK type flip-flop's logic diagram and create its characteristic table and equation. [**CO3**]

c. Draw the D type flip-flop's logic diagram and create its characteristic table and equation. [**CO3**] 10

7. A burglar alarm for a bank is designed so that it senses four input signal lines. Line A is from the secret control switch, line B is from a pressure sensor under a steal safe in a locked closet, line C is from a battery powered clock, and line D is connected to a switch on the locked closet door. The following conditions produce a logic 1 voltage on each line: 30

A: The control switch is closed.

B: The safe is in its normal position in the closet.

C: The clock is between 1000 and 1400 hours.

D: The closet door is closed.

Write the equations with a proper diagram, when an alarm will be produced if the safe is moved and the control switch is closed, or when the closet is opened after banking hours, or when the closet is opened with the control switch open. [**CO4**]

02/03/25

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-221

Course Title: Computer Algorithms and Complexity

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.

b. Answer ***any FIVE*** questions out of **SIX**.

c. Use different answer scripts for each part.

d. Course Learning outcomes are:

CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.

CLO2: Estimate the performance of various algorithms and data structures.

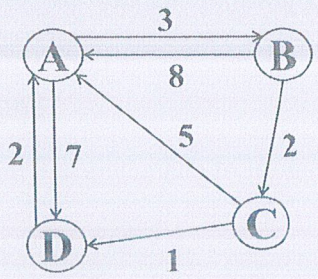
CLO3: Apply important algorithmic design paradigms and methods of analysis for problem solving.

CLO4: Device efficient algorithms in common engineering design situations.

Part-A

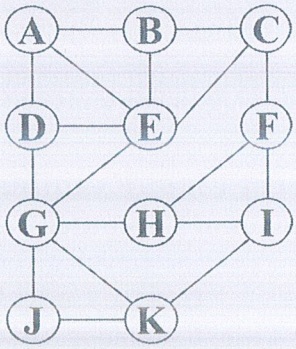
1. a. Apply Dijkstra's algorithm to find the shortest path from node A to node D in a weighted graph with the following edge weights: {(A, B, 6), (A, D, 2), (B, C, 3), (B, D, 4), (C, D, 2), (A, C, 5)}. Show the sequence of nodes visited and the final shortest path distance. **[CLO2→C3]** **15**

b. **15**



Implement the Floyd–Warshall algorithm to find the shortest paths between all pairs of vertices. Start by initializing the distance matrix from the graph's adjacency matrix, then iteratively update it using the Floyd–Warshall update rule. **[CLO3→C3]**

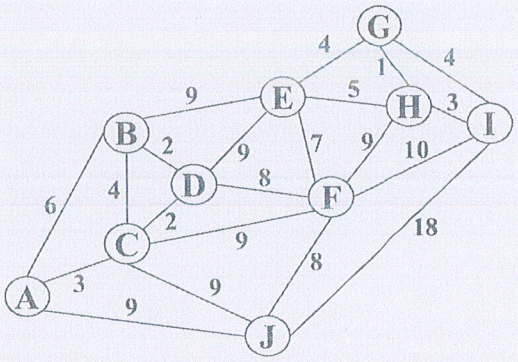
2. a. 20



Perform both Breadth First Search (BFS) and Depth First Search (DFS) in a manner starting from vertex A. List the order in which the vertices are visited during the traversal. **[CLO3→C3]**

b. What is the depth of a tree, and how does it affect the efficiency of different tree traversal techniques? **[CLO2→C2]** 10

3. a. 20



Illustrate the step-by-step execution of Prim's and Kruskal's algorithm on a given weighted graph, indicating how the MST is constructed. **[CLO3→C3]**

b. Can Dijkstra's algorithm handle negative weight edges? Justify your answer. **[CLO2→C4]** 10

Part-B

4. a. Given the following set of characters and their frequencies: 10

Character	Frequency
A	10
B	15
C	30
D	16
E	29

(i) Construct the Huffman tree and determine the Huffman codes for each character.
(ii) Calculate the total number of bits required to encode the given characters using Huffman coding and compare this with fixed-length encoding. **[CLO4→C6]**

b. You are given a knapsack with a weight capacity of 50 units. You have the following items, each with a weight and a value: **10**

Item	Weight (units)	Value (units)
A	10	60
B	20	100
C	30	120
D	15	80
E	25	90
F	5	30
G	12	45
H	18	75

Apply the Greedy Knapsack algorithm to determine the maximum value that can be carried in the knapsack. **[CLO3→C3]**

c. Compare the greedy approach and dynamic programming in terms of problem-solving strategy, efficiency, and application. When should you prefer one over the other? **[CLO3→C4]** **10**

5. a. You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using the Job Sequencing with the Deadline algorithm. **10**

Job	Profit	Deadline
J1	25	3
J2	30	2
J3	15	1
J4	10	4
J5	5	3

Determine the maximum profit by selecting jobs that can be completed before their respective deadlines. **[CLO2→C3]**

b. You are given a bag with a weight capacity of 15 units. You have the following items, each with a weight and a value: **10**

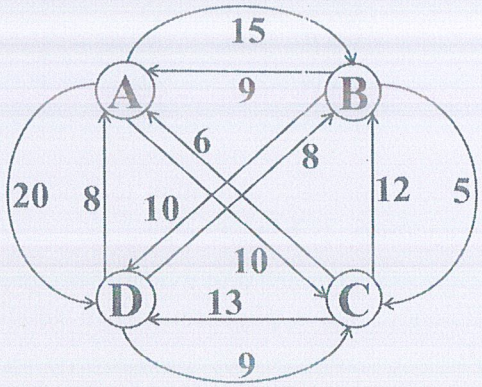
Item	Weight (units)	Value (units)
A	3	4
B	4	5
C	5	7
D	6	8
E	2	3
F	7	9
G	8	11
H	1	2
I	9	13

Apply the 0/1 Knapsack algorithm to determine the maximum value and the items you can carry in the knapsack. **[CLO3→C3]**

c. Provide the pseudocode for the LCS algorithm. How does it work with two sequences to find their longest common subsequence? **[CLO2→C2]** **10**

6. a. Generate the state space tree and apply a backtracking approach to solve the 8-Queen problem. **[CLO4→C6]** **15**

b. Consider the following weighted graph representing a **Travelling Salesman Problem (TSP)**, where the nodes represent cities, and the edges represent travel costs between them: **15**



A salesman starts at city A and must visit all cities exactly once before returning to A. Compute the minimum cost of the tour. **[CLO3→C3]**

06/03/25

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Fall 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***any FIVE*** questions out of **SIX**.
c. Use different answer scripts for each section.
d. Course Learning Outcomes are

CLO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.
CLO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.
CLO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

Part-A

1. a. Construct Newton's backward difference table for given data points and approximate the value of $f(x)$ at x=45. 15

x	10	20	30	40	50
$f(x)$	0.985	0.934	0.866	0.766	0.643

[CLO3->C3, C4, C6]

b. Find the following integral using composite trapezoidal rule for using 2 segments ($k = 2$) and 4 segments ($k = 4$). 15

$$\int_2^4 (x^3 + 2)\ dx$$

[CLO3-> C3, C4, C6]

2. a. Analyze the effect of different initial guesses on the convergence of one-point iteration for $g(x) = 2x^3 - 2x - 5$. **[CLO3-> C3, C4, C6]** 15

b. Use Gauss Quadrature formula, find the solution of $\int_0^1 x^2 e^{-x^2} dx$ for $n = 2$. 15

[CLO2-> C1, C4, C6]

3 a. Using an appropriate Newton's Forword Difference formula, estimate the number of students who obtained marks between 40 and 45 from the following table- 15

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

[CLO3-> C3, C4, C6]

b. Applying Gauss elimination method find the solution for the given equations: 15

$$x + y + z = 2$$
$$x + 2y + 3z = 5$$
$$2x + 3y + 4z = 11$$

[CLO2-> C1, C4, C6]

Part-B

4. a. Justify why iterative methods like Jacobi are preferred for large sparse systems over direct methods like Gauss-Jordan. **[CLO3-> C3, C4, C6]** 15

b. Apply the Jacobi method for two iterations to solve: 15

$$10x + 2y + z = 9$$
$$2x + 10y + 3z = 19$$
$$z + 3x + 10y = 25$$

[CLO2-> C1, C4, C6]

5 a. Approximate the solution of $y' = 3x^2$, $y\,(1) = 1$ using Taylor's series method using first four terms. Approximate the value of $y\,(2)$. **[CLO2-> C1, C4, C6]** 15

b. Estimate the integral value of following function from $x = 1.2$ to 2.4 using Simpson's 1/3 rule. 15

x	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6
$f(x)$	1.53	2.25	3.18	4.32	5.67	7.23	8.98	10.94	13.08

[CLO2-> C1, C4, C6]

6 a. Solve the following set of equations using Gauss Siedal method. 15

$$x + 2y + 3z = 4$$
$$6x + 4y + 5z = 16$$
$$5x + 2y + 3z = 12$$

[CLO3-> C3, C4, C6]

b. Solve the following differential equation for $0 \leq x \leq 1$ taking $h = 0.5$ using Runge Kutta 4th order method. 15

$$y'(x) + y = 3x \text{ with } y(0) = 2.$$

[CLO3-> C3, C4, C6]

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বিএআইইউএসটি) কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) CUMILLA

Term Final Examination, Fall 2024

Department of Computer Science and Engineering
Level-2, Term-II
Course Code: MATH 221
Course Title: MATH IV (Probability and Statistics)
Credit Hour: 3.00

Notes: **Time : 3hrs**

a. Each question carries 30 marks. **Marks : 150**

b. Answer any five (05) questions. **Either three from Part-A and two from Part-B, or two from Part-A and three from Part-B.**

Part-A

1. a. A professor teaches two large sections of introductory marketing and randomly selects a sample of test scores from both sections. **Compute** the variance and standard deviation for each sample: 12

Section 1	50	60	70	80	90
Section 2	72	68	70	74	66

b. The following data relate to the marks obtained by nine students in a class test: 10

Data set	3	4	4	5	5	7	7	7	12

Estimate the mean deviation about mean.

c. The owners are considering purchasing shares of stock A or shares of stock B, both listed on the New York Exchange. From the closing date prices of both stocks over the last several months the means and standard deviations were found to be considerably different, with $\bar{x}_A = \$4.00$, $\bar{x}_B = \$80.00$, $S_A = \$2.00$, and $S_B = \$8.00$. **Find** the coefficient of variation of these two stocks. 8

2. a. **Compute** the coefficient of mean deviation from median for the following distribution table: 13

Class interval	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Frequency	3	1	8	10	7	7	4

b. The following are the marks of 100 students in an examination. **Find** Quartile deviation and coefficient of Quartile deviation. 10

Marks	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Students	4	12	18	28	19	14	5

c. **Interpret** the Range and the coefficient of Range from the following distribution table: 7

Scores	40-45	34-39	28-33	22-27	16-21	10-15
Frequency	3	1	4	5	2	5

3. a. The following data relate to advertising expenditure and sales of a firm: 15

Advertising Expenditure: x	14	19	24	21	26	22	15	20	19
Sales (in lakh tk): y	31	36	48	37	50	45	33	41	39

Derive the regression equation of sales on advertising expenditure and interpret a and b.

b. **Obtain** a multiple linear regression equation having the form $y = b_0 + b_1x_1 + b_2x_2$ to the given data: 15

Y	12	14	`18	25	30
x_1	3	5	8	4	7
x_2	6	4	9	8	1

Part-B

1. a. **Compute** Karl Pearson correlation coefficient for the following pair of set and draw a scatter plot. Also comment on the nature of the following data. 13

x	1	2	3	4	5
y	10	8	6	4	2

b. Ten students appearing at an examination were evaluated by two independent examiners out of 100 marks. We wish to determine whether the marks given by these examiners are correlated. Table below shows these marks. 10

Examiner 1	65	70	76	75	80	78	83	84	85	90
Examiner 2	30	25	35	40	38	42	48	50	55	45

Hence **illustrate** the rank correlation co-efficient.

c. Machine produces parts with mean length of 4.125 mm. Customer randomly choose 20 parts and find sample length as 4.120 mm. Suppose sample standard deviation is 0.008 mm. Use $\alpha = 0.10$ and **determine** whether the mean length has decreased? 07

2. a. Two random samples were drawn from the two normal populations and their values are: 13

A	16	17	25	26	32	34	38	40	42		
B	14	16	24	28	32	35	37	42	43	45	47

Determine whether the two populations have the same variance at 5% level of significance.

b. The first four raw moments of a distribution are 2, 136, 320 and 40,000. **Obtain** the Karl Pearson's coefficient of skewness and kurtosis. Comment on the nature of the distribution. 10

c. Karl Pearson's coefficient of skewness is 1.28, its mean is 164 and mode 100, **demonstrate** the standard deviation. 07

3. a. To study the performance of three detergents and three different water temperatures, the following whiteness readings were obtained with specially designed equipment. 20

Water temperature	A	B	C
Cold water	47	45	50
Warm water	39	42	52
Hot water	44	36	48

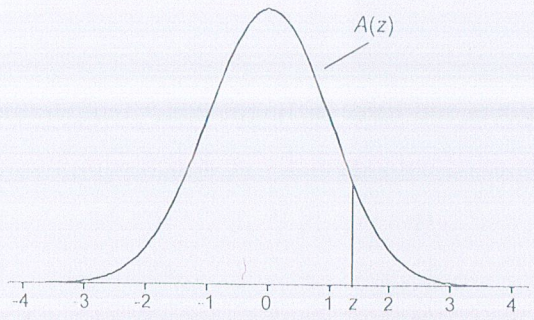
Using the long method, **determine** the test at the 5% significance level whether there is a significance difference between the detergent and water temperature.

b. The following are the marks of 150 students in an examination. **Calculate** Karl Pearson's coefficient of skewness. 10

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students	10	40	20	0	10	40	16	14

TABLE A.1

Cumulative Standardized Normal Distribution



$A(z)$ is the integral of the standardized normal distribution from $-\infty$ to z (in other words, the area under the curve to the left of z). It gives the probability of a normal random variable not being more than z standard deviations above its mean. Values of z of particular importance:

z	$A(z)$	
1.645	0.9500	Lower limit of right 5% tail
1.960	0.9750	Lower limit of right 2.5% tail
2.326	0.9900	Lower limit of right 1% tail
2.576	0.9950	Lower limit of right 0.5% tail
3.090	0.9990	Lower limit of right 0.1% tail
3.291	0.9995	Lower limit of right 0.05% tail

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							

TABLE A.2

t Distribution: Critical Values of *t*

Degrees of freedom	*Two-tailed test:* / *One-tailed test:*	*Significance level* 10% / 5%	5% / 2.5%	2% / 1%	1% / 0.5%	0.2% / 0.1%	0.1% / 0.05%
1		6.314	12.706	31.821	63.657	318.309	636.619
2		2.920	4.303	6.965	9.925	22.327	31.599
3		2.353	3.182	4.541	5.841	10.215	12.924
4		2.132	2.776	3.747	4.604	7.173	8.610
5		2.015	2.571	3.365	4.032	5.893	6.869
6		1.943	2.447	3.143	3.707	5.208	5.959
7		1.894	2.365	2.998	3.499	4.785	5.408
8		1.860	2.306	2.896	3.355	4.501	5.041
9		1.833	2.262	2.821	3.250	4.297	4.781
10		1.812	2.228	2.764	3.169	4.144	4.587
11		1.796	2.201	2.718	3.106	4.025	4.437
12		1.782	2.179	2.681	3.055	3.930	4.318
13		1.771	2.160	2.650	3.012	3.852	4.221
14		1.761	2.145	2.624	2.977	3.787	4.140
15		1.753	2.131	2.602	2.947	3.733	4.073
16		1.746	2.120	2.583	2.921	3.686	4.015
17		1.740	2.110	2.567	2.898	3.646	3.965
18		1.734	2.101	2.552	2.878	3.610	3.922
19		1.729	2.093	2.539	2.861	3.579	3.883
20		1.725	2.086	2.528	2.845	3.552	3.850
21		1.721	2.080	2.518	2.831	3.527	3.819
22		1.717	2.074	2.508	2.819	3.505	3.792
23		1.714	2.069	2.500	2.807	3.485	3.768
24		1.711	2.064	2.492	2.797	3.467	3.745
25		1.708	2.060	2.485	2.787	3.450	3.725
26		1.706	2.056	2.479	2.779	3.435	3.707
27		1.703	2.052	2.473	2.771	3.421	3.690
28		1.701	2.048	2.467	2.763	3.408	3.674
29		1.699	2.045	2.462	2.756	3.396	3.659
30		1.697	2.042	2.457	2.750	3.385	3.646
32		1.694	2.037	2.449	2.738	3.365	3.622
34		1.691	2.032	2.441	2.728	3.348	3.601
36		1.688	2.028	2.434	2.719	3.333	3.582
38		1.686	2.024	2.429	2.712	3.319	3.566
40		1.684	2.021	2.423	2.704	3.307	3.551
42		1.682	2.018	2.418	2.698	3.296	3.538
44		1.680	2.015	2.414	2.692	3.286	3.526
46		1.679	2.013	2.410	2.687	3.277	3.515
48		1.677	2.011	2.407	2.682	3.269	3.505
50		1.676	2.009	2.403	2.678	3.261	3.496
60		1.671	2.000	2.390	2.660	3.232	3.460
70		1.667	1.994	2.381	2.648	3.211	3.435
80		1.664	1.990	2.374	2.639	3.195	3.416
90		1.662	1.987	2.368	2.632	3.183	3.402
100		1.660	1.984	2.364	2.626	3.174	3.390
120		1.658	1.980	2.358	2.617	3.160	3.373
150		1.655	1.976	2.351	2.609	3.145	3.357
200		1.653	1.972	2.345	2.601	3.131	3.340
300		1.650	1.968	2.339	2.592	3.118	3.323
400		1.649	1.966	2.336	2.588	3.111	3.315
500		1.648	1.965	2.334	2.586	3.107	3.310
600		1.647	1.964	2.333	2.584	3.104	3.307
∞		1.645	1.960	2.326	2.576	3.090	3.291

TABLE A.3

F Distribution: Critical Values of *F* (5% significance level)

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.36	246.46	247.32	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.42	19.43	19.44	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.71	8.69	8.67	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.87	5.84	5.82	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.64	4.60	4.58	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.96	3.92	3.90	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.53	3.49	3.47	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.24	3.20	3.17	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.03	2.99	2.96	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.86	2.83	2.80	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.74	2.70	2.67	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.64	2.60	2.57	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.55	2.51	2.48	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.48	2.44	2.41	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.42	2.38	2.35	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.37	2.33	2.30	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.33	2.29	2.26	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.29	2.25	2.22	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.26	2.21	2.18	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.22	2.18	2.15	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.20	2.16	2.12	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.17	2.13	2.10	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.15	2.11	2.08	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.13	2.09	2.05	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.11	2.07	2.04	2.01
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.09	2.05	2.02	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.08	2.04	2.00	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.06	2.02	1.99	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.05	2.01	1.97	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.04	1.99	1.96	1.93
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11	2.04	1.99	1.94	1.91	1.88
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.95	1.90	1.87	1.84
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.95	1.89	1.85	1.81	1.78
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.86	1.82	1.78	1.75
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97	1.89	1.84	1.79	1.75	1.72
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.88	1.82	1.77	1.73	1.70
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94	1.86	1.80	1.76	1.72	1.69
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.85	1.79	1.75	1.71	1.68
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.78	1.73	1.69	1.66
150	3.90	3.06	2.66	2.43	2.27	2.16	2.07	2.00	1.94	1.89	1.82	1.76	1.71	1.67	1.64
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88	1.80	1.74	1.69	1.66	1.62
250	3.88	3.03	2.64	2.41	2.25	2.13	2.05	1.98	1.92	1.87	1.79	1.73	1.68	1.65	1.61
300	3.87	3.03	2.63	2.40	2.24	2.13	2.04	1.97	1.91	1.86	1.78	1.72	1.68	1.64	1.61
400	3.86	3.02	2.63	2.39	2.24	2.12	2.03	1.96	1.90	1.85	1.78	1.72	1.67	1.63	1.60
500	3.86	3.01	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85	1.77	1.71	1.66	1.62	1.59
600	3.86	3.01	2.62	2.39	2.23	2.11	2.02	1.95	1.90	1.85	1.77	1.71	1.66	1.62	1.59
750	3.85	3.01	2.62	2.38	2.23	2.11	2.02	1.95	1.89	1.84	1.77	1.70	1.66	1.62	1.58
1000	3.85	3.00	2.61	2.38	2.22	2.11	2.02	1.95	1.89	1.84	1.76	1.70	1.65	1.61	1.58

TABLE A.3 (continued)

F Distribution: Critical Values of *F* (5% significance level)

ν_2 \ ν_1	25	30	35	40	50	60	75	100	150	200
1	249.26	250.10	250.69	251.14	251.77	252.20	252.62	253.04	253.46	253.68
2	19.46	19.46	19.47	19.47	19.48	19.48	19.48	19.49	19.49	19.49
3	8.63	8.62	8.60	8.59	8.58	8.57	8.56	8.55	8.54	8.54
4	5.77	5.75	5.73	5.72	5.70	5.69	5.68	5.66	5.65	5.65
5	4.52	4.50	4.48	4.46	4.44	4.43	4.42	4.41	4.39	4.39
6	3.83	3.81	3.79	3.77	3.75	3.74	3.73	3.71	3.70	3.69
7	3.40	3.38	3.36	3.34	3.32	3.30	3.29	3.27	3.26	3.25
8	3.11	3.08	3.06	3.04	3.02	3.01	2.99	2.97	2.96	2.95
9	2.89	2.86	2.84	2.83	2.80	2.79	2.77	2.76	2.74	2.73
10	2.73	2.70	2.68	2.66	2.64	2.62	2.60	2.59	2.57	2.56
11	2.60	2.57	2.55	2.53	2.51	2.49	2.47	2.46	2.44	2.43
12	2.50	2.47	2.44	2.43	2.40	2.38	2.37	2.35	2.33	2.32
13	2.41	2.38	2.36	2.34	2.31	2.30	2.28	2.26	2.24	2.23
14	2.34	2.31	2.28	2.27	2.24	2.22	2.21	2.19	2.17	2.16
15	2.28	2.25	2.22	2.20	2.18	2.16	2.14	2.12	2.10	2.10
16	2.23	2.19	2.17	2.15	2.12	2.11	2.09	2.07	2.05	2.04
17	2.18	2.15	2.12	2.10	2.08	2.06	2.04	2.02	2.00	1.99
18	2.14	2.11	2.08	2.06	2.04	2.02	2.00	1.98	1.96	1.95
19	2.11	2.07	2.05	2.03	2.00	1.98	1.96	1.94	1.92	1.91
20	2.07	2.04	2.01	1.99	1.97	1.95	1.93	1.91	1.89	1.88
21	2.05	2.01	1.98	1.96	1.94	1.92	1.90	1.88	1.86	1.84
22	2.02	1.98	1.96	1.94	1.91	1.89	1.87	1.85	1.83	1.82
23	2.00	1.96	1.93	1.91	1.88	1.86	1.84	1.82	1.80	1.79
24	1.97	1.94	1.91	1.89	1.86	1.84	1.82	1.80	1.78	1.77
25	1.96	1.92	1.89	1.87	1.84	1.82	1.80	1.78	1.76	1.75
26	1.94	1.90	1.87	1.85	1.82	1.80	1.78	1.76	1.74	1.73
27	1.92	1.88	1.86	1.84	1.81	1.79	1.76	1.74	1.72	1.71
28	1.91	1.87	1.84	1.82	1.79	1.77	1.75	1.73	1.70	1.69
29	1.89	1.85	1.83	1.81	1.77	1.75	1.73	1.71	1.69	1.67
30	1.88	1.84	1.81	1.79	1.76	1.74	1.72	1.70	1.67	1.66
35	1.82	1.79	1.76	1.74	1.70	1.68	1.66	1.63	1.61	1.60
40	1.78	1.74	1.72	1.69	1.66	1.64	1.61	1.59	1.56	1.55
50	1.73	1.69	1.66	1.63	1.60	1.58	1.55	1.52	1.50	1.48
60	1.69	1.65	1.62	1.59	1.56	1.53	1.51	1.48	1.45	1.44
70	1.66	1.62	1.59	1.57	1.53	1.50	1.48	1.45	1.42	1.40
80	1.64	1.60	1.57	1.54	1.51	1.48	1.45	1.43	1.39	1.38
90	1.63	1.59	1.55	1.53	1.49	1.46	1.44	1.41	1.38	1.36
100	1.62	1.57	1.54	1.52	1.48	1.45	1.42	1.39	1.36	1.34
120	1.60	1.55	1.52	1.50	1.46	1.43	1.40	1.37	1.33	1.32
150	1.58	1.54	1.50	1.48	1.44	1.41	1.38	1.34	1.31	1.29
200	1.56	1.52	1.48	1.46	1.41	1.39	1.35	1.32	1.28	1.26
250	1.55	1.50	1.47	1.44	1.40	1.37	1.34	1.31	1.27	1.25
300	1.54	1.50	1.46	1.43	1.39	1.36	1.33	1.30	1.26	1.23
400	1.53	1.49	1.45	1.42	1.38	1.35	1.32	1.28	1.24	1.22
500	1.53	1.48	1.45	1.42	1.38	1.35	1.31	1.28	1.23	1.21
600	1.52	1.48	1.44	1.41	1.37	1.34	1.31	1.27	1.23	1.20
750	1.52	1.47	1.44	1.41	1.37	1.34	1.30	1.26	1.22	1.20
1000	1.52	1.47	1.43	1.41	1.36	1.33	1.30	1.26	1.22	1.19